



Sales Performance Analysis of a Retail Chain Using Business Intelligence Tools

Capstone Project – Business Intelligence

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Introduction

Business Scenario

The retail chain wants to understand its sales performance across different cities, products, and time periods. The company collected transactional sales data but needed analysis to identify trends, improve decision-making, and optimize business operations.

- **Purpose of the Analysis**
- The purpose of this analysis is to:
- Clean and prepare raw sales data
- Analyze sales performance using analytical tools
- Identify patterns and trends
- Visualize business insights
- Provide recommendations to improve sales and inventory management





Methodology

-
- The project followed a structured analytical approach beginning with data cleaning and preparation, where missing values, formatting issues, and inconsistencies were addressed to ensure data reliability.
 - Next, exploratory data analysis was performed using Pivot Tables in Excel to summarize sales by date, city, and store type. Statistical analysis, including regression analysis, was conducted to evaluate the relationship between sales and time and to identify trends.
 - Data visualization techniques were then applied in Excel to convert raw data into meaningful charts, followed by the development of Tableau visualizations and interactive dashboards. Each module built upon the previous one to ensure a logical and systematic analytical workflow that supports business decision-making.

Data Details

The dataset consisted of retail sales transaction records provided in CSV format. It included structured information such as:

- Order ID
- Order Date
- City
- Product Name
- Category
- Quantity Sold
- Sales Amount

The dataset was imported into Microsoft Excel for initial cleaning and later into PostgreSQL for advanced querying and analysis.

Module 1: Data Cleaning and Preparation Using Excel

- Removed duplicate records.
- Standardized date and text formats.
- Trimmed extra spaces and corrected inconsistencies.
- Ensured proper product hierarchy structure.

This step ensured accuracy, reliability, and consistency in the dataset before performing analysis.

Module 2: Data Querying and Analysis Using PostgreSQL

- Imported cleaned data into the BICapstone PostgreSQL database.
- Verified data integrity using SELECT queries.
- Performed aggregation using SQL functions.
- Conducted sales performance analysis using GROUP BY and ROLLUP.

Using PostgreSQL allowed efficient handling of large datasets and advanced querying capabilities.

Module 3: Data Visualization and Statistical Analysis

- Created pivot tables for summary analysis.
- Developed line charts, bar charts, sunburst charts, and scatter plots.
- Performed regression analysis to study relationships between variables.

Visualizations and statistical methods enhanced understanding of sales patterns and trends.

Module 4: Data Visualization and Dashboards Using Tableau

- Created interactive visualizations including line charts, bar charts, scatter plots, and packed bubbles.
- Designed a dashboard integrating multiple worksheets.
- Implemented filter dashboard actions for dynamic data exploration.

Tableau enabled effective communication of insights through interactive dashboards.



Results

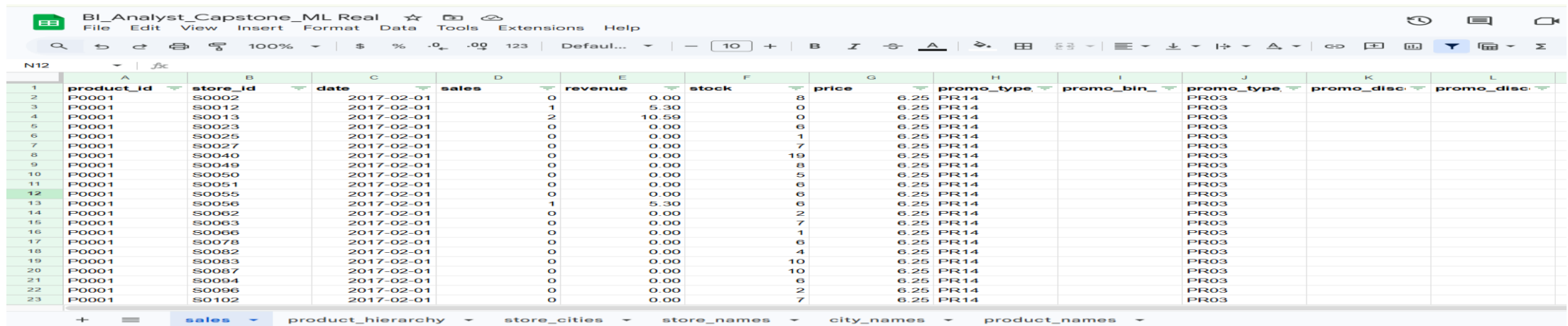
Module 1, Lesson 1: Data Cleaning and Preparation

Data Cleaning Results

- The dataset was cleaned by removing duplicate records, correcting formatting inconsistencies, trimming extra spaces, and standardizing date and text formats. Missing or inconsistent values were corrected to ensure accuracy.

Significance of Data Cleaning

- Data cleaning is critical because it ensures reliability and accuracy in analysis. Clean data prevents incorrect calculations, reduces errors in reporting, and improves confidence in business decisions derived from the analysis.



The screenshot displays a spreadsheet application window titled "BI_Analyst_Capstone_ML Real". The spreadsheet contains a table with 12 columns (A-L) and 23 rows of data. The columns are: product_id, store_id, date, sales, revenue, stock, price, promo_type, promo_bin, promo_type, promo_disc, and promo_disc. The data represents sales records for various products and stores over time, with standardized values for each field.

	A	B	C	D	E	F	G	H	I	J	K	L
	product_id	store_id	date	sales	revenue	stock	price	promo_type	promo_bin	promo_type	promo_disc	promo_disc
1	P0001	S0002	2017-02-01	0	0.00	8	6.25	PR14		PR03		
2	P0001	S0012	2017-02-01	1	5.30	0	6.25	PR14		PR03		
3	P0001	S0013	2017-02-01	2	10.50	0	6.25	PR14		PR03		
4	P0001	S0023	2017-02-01	0	0.00	6	6.25	PR14		PR03		
5	P0001	S0025	2017-02-01	0	0.00	1	6.25	PR14		PR03		
6	P0001	S0027	2017-02-01	0	0.00	7	6.25	PR14		PR03		
7	P0001	S0040	2017-02-01	0	0.00	19	6.25	PR14		PR03		
8	P0001	S0049	2017-02-01	0	0.00	8	6.25	PR14		PR03		
9	P0001	S0050	2017-02-01	0	0.00	5	6.25	PR14		PR03		
10	P0001	S0051	2017-02-01	0	0.00	6	6.25	PR14		PR03		
11	P0001	S0055	2017-02-01	0	0.00	6	6.25	PR14		PR03		
12	P0001	S0056	2017-02-01	1	5.30	6	6.25	PR14		PR03		
13	P0001	S0062	2017-02-01	0	0.00	2	6.25	PR14		PR03		
14	P0001	S0063	2017-02-01	0	0.00	7	6.25	PR14		PR03		
15	P0001	S0066	2017-02-01	0	0.00	1	6.25	PR14		PR03		
16	P0001	S0076	2017-02-01	0	0.00	6	6.25	PR14		PR03		
17	P0001	S0082	2017-02-01	0	0.00	4	6.25	PR14		PR03		
18	P0001	S0083	2017-02-01	0	0.00	10	6.25	PR14		PR03		
19	P0001	S0087	2017-02-01	0	0.00	10	6.25	PR14		PR03		
20	P0001	S0094	2017-02-01	0	0.00	6	6.25	PR14		PR03		
21	P0001	S0096	2017-02-01	0	0.00	2	6.25	PR14		PR03		
22	P0001	S0102	2017-02-01	0	0.00	7	6.25	PR14		PR03		
23	P0001	S0102	2017-02-01	0	0.00	7	6.25	PR14		PR03		

Module 1, Lesson 2: Data Analysis Using Pivot Tables

Sales Performance Analysis

- The pivot table summarized total sales across all transactions. This helped identify overall revenue performance and detect patterns in sales distribution.

Business Significance

- Understanding total sales performance provides insight into business growth, revenue trends, and overall financial health.

This screenshot shows a PivotTable summarizing sales data by store. The PivotTable is located in the range E7:K23. The data source is a table with columns: store_name, SUM of sales, SUM of revenue, SUM of stock, and AVERAGE of sales. The PivotTable is set to show the grand total for all stores, with the value for the average of sales highlighted as 0.4884026258.

	A	B	C	D	E
1	store_name	SUM of sales	SUM of revenue	SUM of stock	AVERAGE of sales
2	AGD Centrum	129	303.74	4683	0.3200992556
3	AGD Expert	29	252.2	5812	0.06401766004
4	Appliance Cent	64	122.37	3400	0.2976744186
5	Appliance Direct	89	197.62	6406	0.2299741602
6	Appliance Parad	114	253.02	2223	0.5135135135
7	Appliance World	223.2	755.83	5928.695	0.4884026258
8	Bimeks	158	313.17	4894	0.4213333333
9	Block	945.507	2864.18	16626.721	1.260676
10	Boulangier (Natic	105	275.34	6547	0.2071005917
11	Casa da Tecnolo	36	72.14	4050	0.09
12	Casa Digital	200	695.1	17073	0.180342651
13	Casa Tech	323	897.21	10981	0.569664903
14	Cecchini	81	152.3	2484	0.4879518072
15	Centro delloElett	144	363.46	4630	0.3486662809
16	Comet	91.58	276.27	4859.82	0.2266831683
17	Conrad Electron	39	73.59	2673	0.1764705882
18	Coolblue	402.895	1289.11	5383.29	0.9008936689
19	Cosmos Sport	72	197.89	5471	0.1431411531
20	Currys	849.372	1673.98	35077.67	0.9074487179
21	Currys (National	83	194.22	7415	0.1443478261
22	Darty	199.18	414.75	6727.575	0.4274248927
23	ty (National C	90	122.12	2964	0.5325443787

This screenshot shows a PivotTable summarizing sales data by city. The PivotTable is located in the range A1:J24. The data source is a table with columns: city_name, SUM of sales, SUM of revenue, and AVERAGE of sales. The PivotTable is set to show the grand total for all cities, with the value for the average of sales highlighted as 0.2696849758.

	A	B	C	D
1	city_name	SUM of sales	SUM of revenue	AVERAGE of sales
3	Barcelona	108.84	209.53	0.2513625866
4	Belgrade	598.725	1409.15	0.3739693941
5	Berlin	505.508	1119.53	0.6431399491
6	Brussels	337	652.34	0.3820861678
7	Budapest	849.372	1673.98	0.9074487179
8	Copenhagen	275	596.66	0.353470437
9	Dublin	408.83	843.64	0.2495909646
10	Edinburgh	334.39	875.68	0.4797560976
11	Florence	467.725	1381.19	0.2831265133
12	Frankfurt	72	197.89	0.1431411531
13	Geneva	83	197.59	0.1680161943
14	Helsinki	2387.45	5273.97	0.6294358028
15	Istanbul	326.47	796.99	0.493903177
16	Krakow	165	427.57	0.3210116732
17	Kyiv	152	364	0.3828715365
18	Lisbon	198	494.08	0.3193548387
19	London	9507.201	32143.98	0.761672889
20	Luxembourg	204	683.62	0.1673502871
21	Madrid	281.593	856.52	0.3700302234
22	Milan	253.405	794.55	0.3286705577
23	Munich	306.254	882.5	0.4277290503
24	o	166.935	403.07	0.2696849758

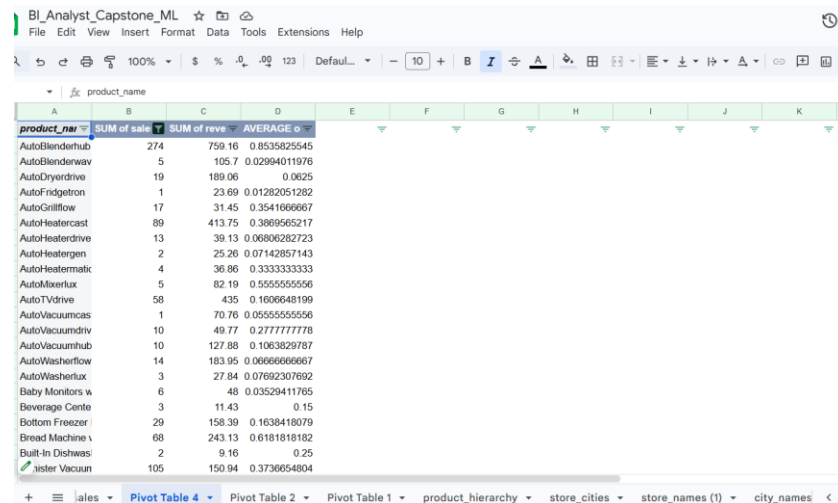
Product Performance

Product Performance Analysis

- The pivot table identified best-selling and least-performing products based on sales value and quantity sold.

Business Significance

- This insight supports stock optimization, helps reduce overstocking of slow-moving items, and improves profitability by promoting high-performing products.



product_name	SUM of sale	SUM of revs	AVERAGE o
AutoBlenderhub	274	759.16	0.8535825545
AutoBlenderwav	5	105.7	0.02904011976
AutoDryerdrive	19	189.06	0.0625
AutoFridgetron	1	23.69	0.01282051282
AutoGrillflow	17	31.45	0.3541606667
AutoHeatercast	89	413.75	0.3869565217
AutoHeaterdrive	13	39.13	0.06806282723
AutoHeatergen	2	25.26	0.07142857143
AutoHeatermatic	4	36.86	0.3333333333
AutoMixerlux	5	82.19	0.5555555556
AutoTVdrive	58	435	0.1666648199
AutoVacuumcas	1	70.79	0.05555555556
AutoVacuumdriv	10	49.77	0.2777777778
AutoVacuumhub	10	127.88	0.1063829787
AutoWasherflow	14	183.95	0.0666666667
AutoWasherlux	3	27.84	0.07692307692
Baby Monitors v	6	49	0.03529411765
Beverage Cente	3	11.43	0.15
Bottom Freezer	29	158.39	0.1638418079
Bread Machine v	68	243.13	0.6181818182
Built-In Dishwas	2	9.16	0.25
hister Vacuum	105	150.94	0.3736654804

Module 2, Lesson 1: Data Querying Using PostgreSQL

Database Setup and Querying

- The cleaned dataset was imported into the BICapstone PostgreSQL database. SQL queries were written to verify data integrity and perform sales performance analysis using aggregation functions.

Importance of Database Setup

- Setting up the database allows efficient handling of large datasets, faster querying, and structured storage. It ensures scalability and enables advanced analysis techniques beyond spreadsheet limitations.

Module 2, Lesson 2: Data Analysis Using PostgreSQL

Advanced SQL Analysis

- Data cubes were created using ROLLUP to generate hierarchical summaries across cities, categories, and time periods. Trend analysis queries identified changes in sales over time and across regions.

Business Impact

- These insights enable better demand forecasting, improved inventory planning, and region-specific sales strategies. Hierarchical summaries support high-level managerial reporting.

Module 3, Lesson 1: Data Visualization Using Excel

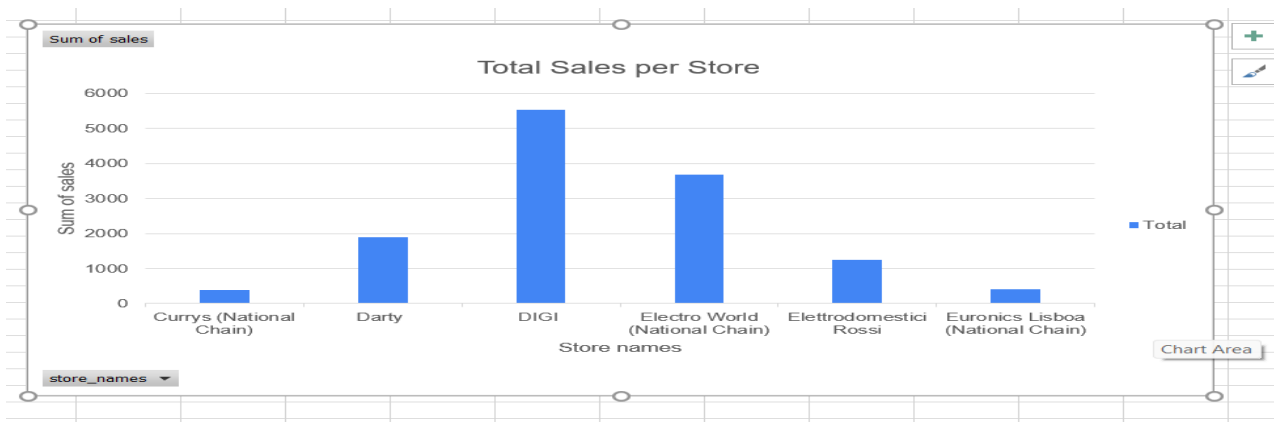
Line Chart

→ Shows sales trends over time.



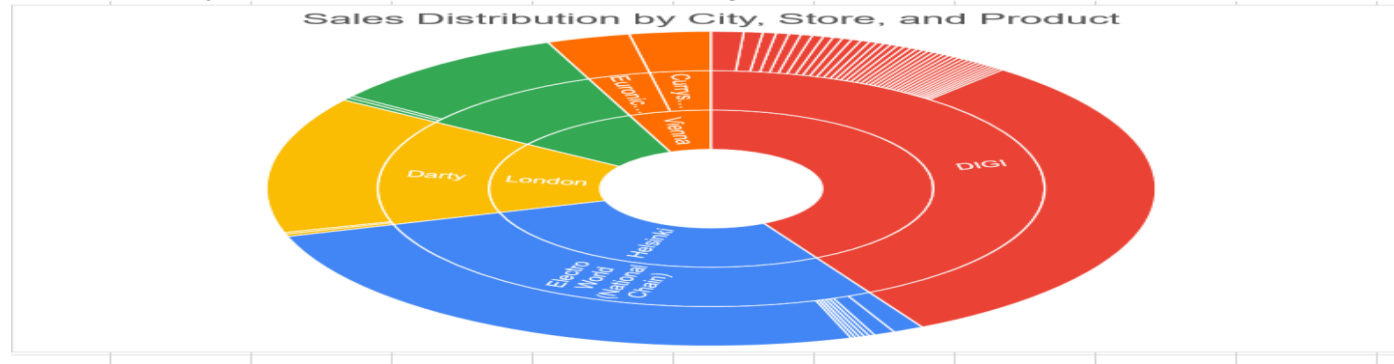
Bar Chart

→ Compares sales across cities/products.



Sunburst Chart

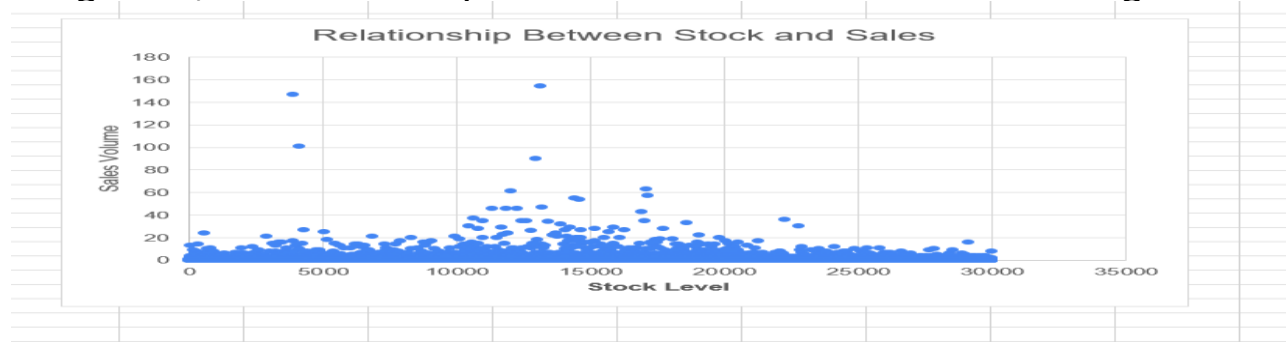
→ Displays hierarchical product categories.



Scatter Plot

→ Shows relationships between sales and revenue.

Together, these charts provide a clear visual understanding of business performance.



29							
	A	B	C	D	E	F	
7	Standard Error	2.805713082					
8	Observations	19999					
9							
10	ANOVA						
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>	
12	Regression	1	1.021877362	1.021877362	0.12981123	0.718631648	
13	Residual	19997	157416.9019	7.872025898			
14	Total	19998	157417.9238				
15							
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>
17	Intercept	-23.18840106	65.8376272	-0.352205905	0.72468756	-152.2355901	
18	date	0.000554764	0.001539758	0.360293254	0.718631649	-0.002463288	
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24	Future Date (Nu	Predicted Sales					
25	43101	0.722482104					
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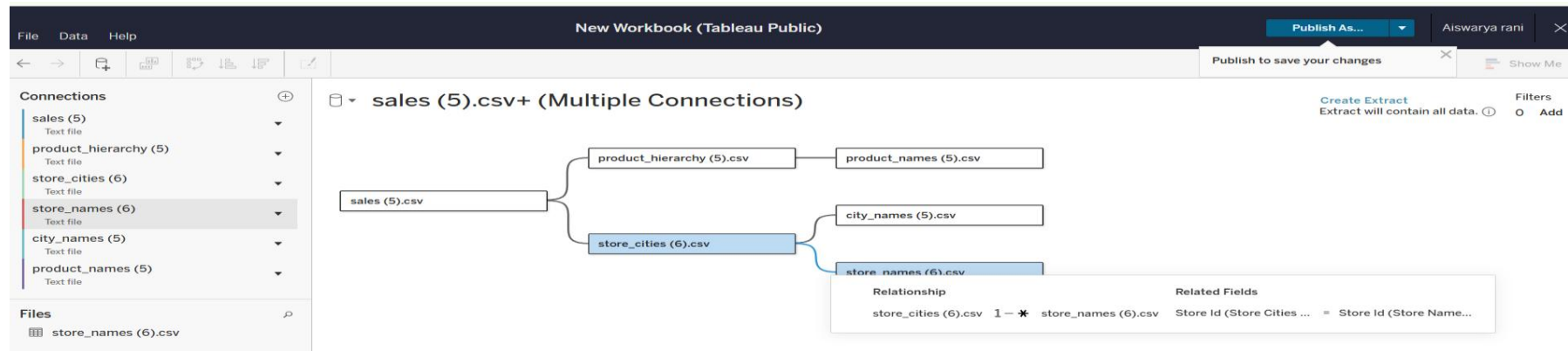
Module 4, Lesson 1: Basic Tableau Visualizations

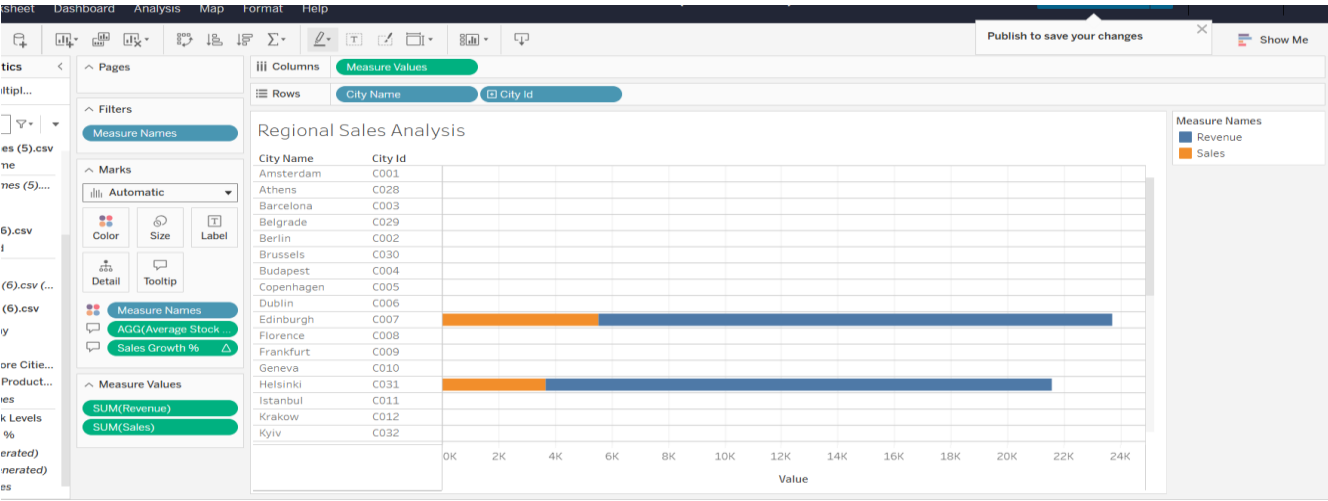
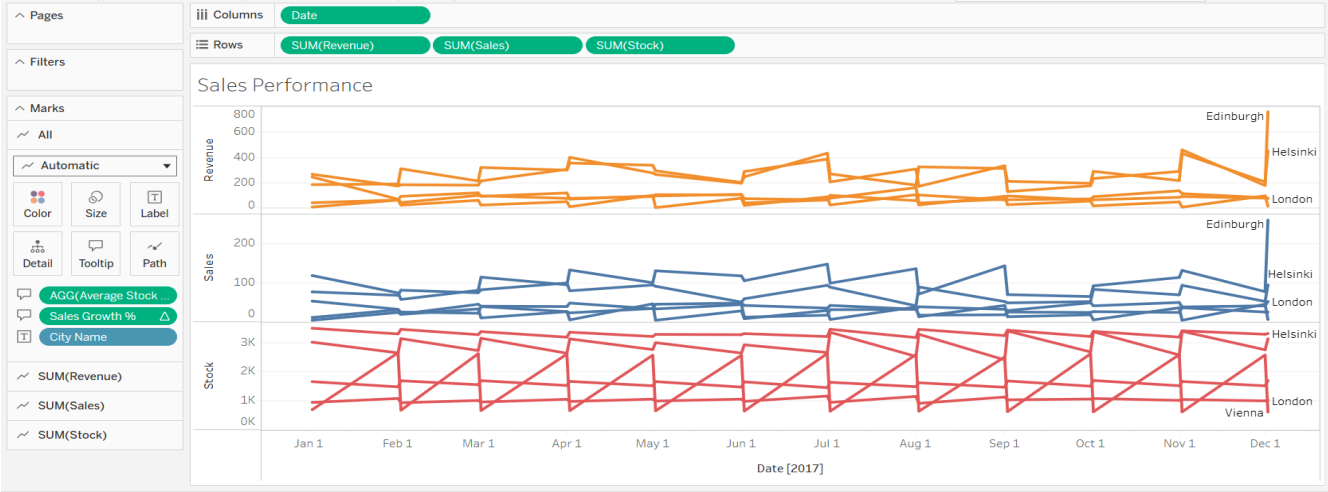
Tableau Visualizations

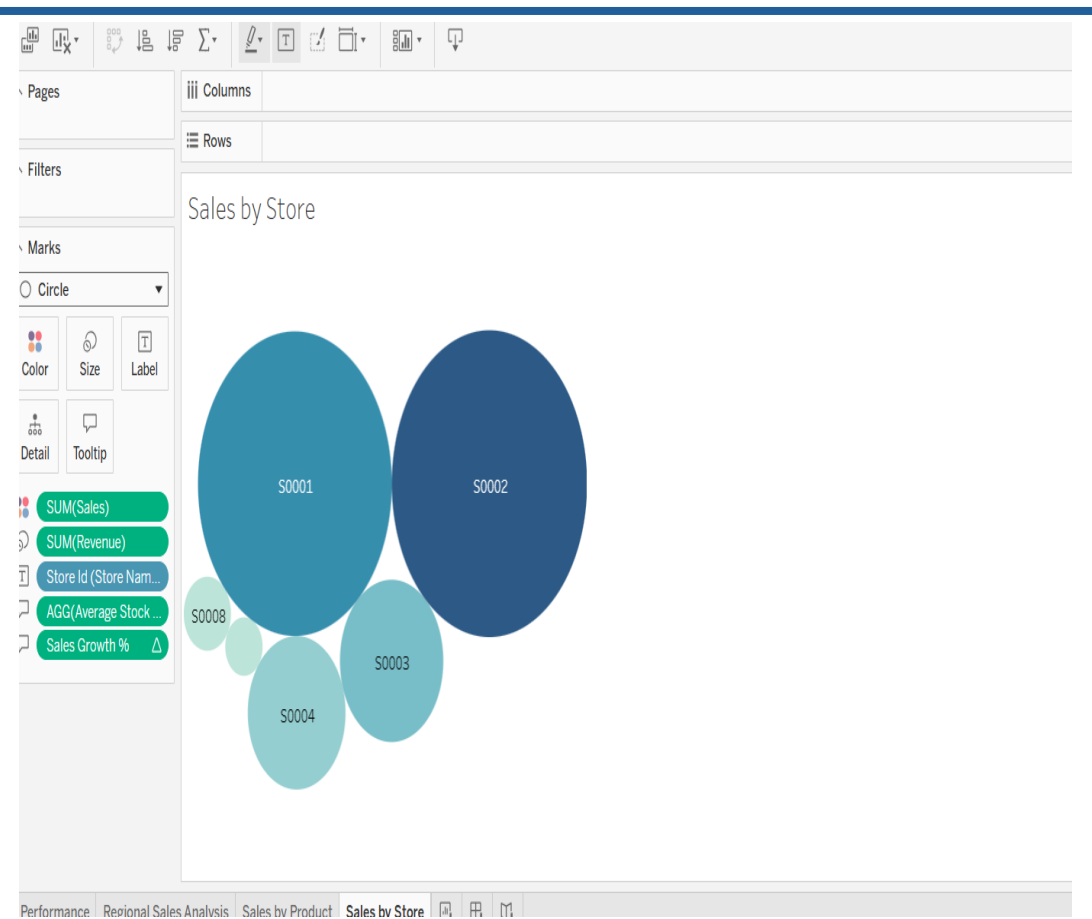
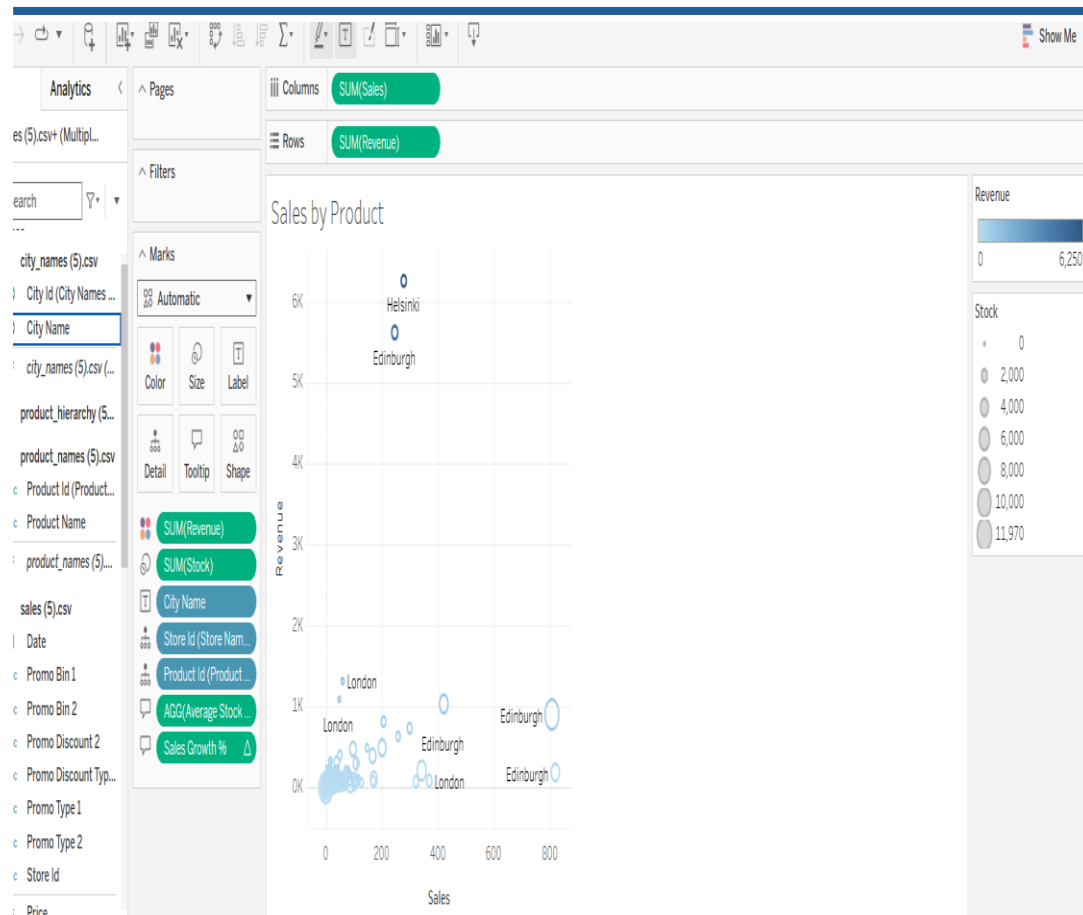
- Interactive visualizations were created using Tableau to analyze sales trends, compare performance across regions, and explore relationships between variables.

Communication Value

- Tableau enhances data storytelling by providing dynamic and interactive visualizations, making insights more accessible to stakeholders.







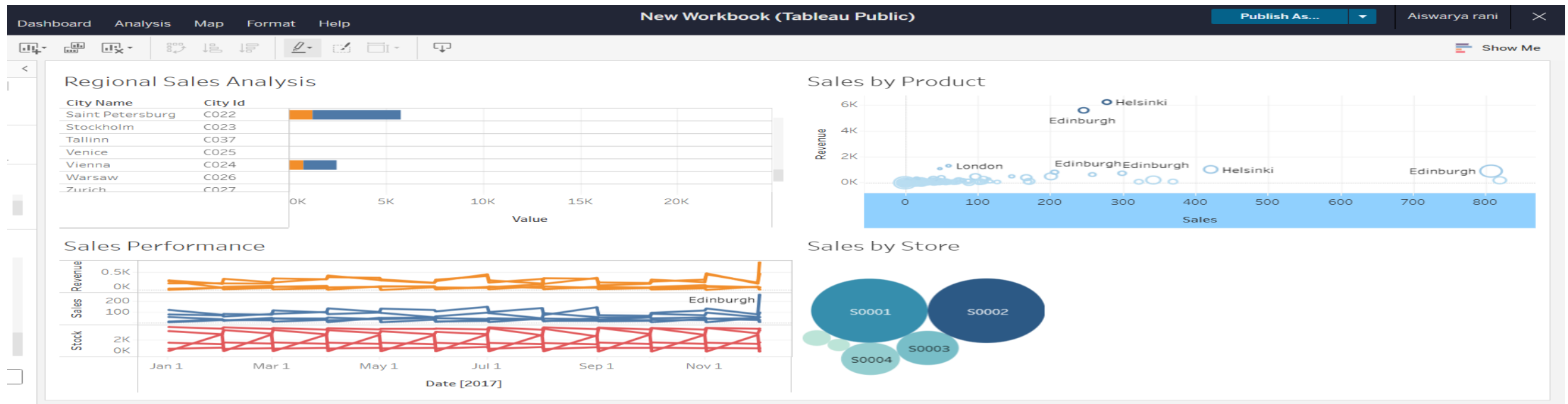
Module 4, Lesson 2: Advanced Visualizations Using Tableau

Dashboard and Filter Implementation

- A comprehensive dashboard was created by integrating multiple worksheets. Filter actions were implemented to allow dynamic exploration of data by city, category, and time period.

Decision-Making Support

- Interactive dashboards enable management to quickly analyze performance metrics, identify trends, and make data-driven strategic decisions.





Discussion

Insights and Recommendations

Key Insights

- Sales performance varies significantly across cities, with certain regions contributing a higher share of total revenue.
- A small number of products generate a large proportion of total sales.
- Sales trends indicate seasonal fluctuations and growth patterns over time.
- Statistical analysis suggests a measurable relationship between selected variables affecting sales.

Business Implications

- High-performing regions should receive focused marketing and inventory support.
- Underperforming cities require targeted promotional strategies.
- Inventory planning should prioritize high-demand products to avoid stockouts.
- Sales forecasting can be improved using historical trend patterns.



Conclusion

Summary

- This analysis evaluated the retail chain's sales performance using Excel, PostgreSQL, and Tableau. The study identified top-performing cities, high-demand products, and measurable sales trends. Statistical and visual analysis supported data-driven insights.
- Overall, business intelligence tools enabled structured analysis, improved reporting clarity, and provided actionable recommendations to enhance strategic decision-making.

Innovation and Value Addition

- Yes, innovation and value addition have been demonstrated in the presentation. Beyond basic analysis, interactive dashboards were developed to enhance usability and stakeholder engagement.
- The integration of regression analysis adds predictive insight into sales trends, providing forward-looking value rather than just descriptive reporting. Additionally, the structured combination of Excel analysis and Tableau visualization strengthens the overall analytical depth of the project.
- This approach ensures that the presentation not only reports findings but also provides actionable business insights that support strategic decision-making.



Appendix

Appendix – Supporting Analysis

- Additional SQL queries
- Extra pivot tables
- Extra charts not shown earlier